

D4-Equivariant Hybrid Quantum–Classical Learning for Strong Gravitational Lens Classification

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Abstract

Strong gravitational lensing provides a powerful probe of small-scale dark matter structure, but extracting this information requires distinguishing weak morphological perturbations from source structure, instrumental effects, and noise while remaining insensitive to physically irrelevant image orientations. Although group-equivariant classical networks can encode rotational and reflection symmetry, it remains unclear whether a parameterized quantum circuit can provide additional representational capacity beyond a strong symmetry-aware classical model. We hypothesize that a symmetry-preserving quantum circuit can refine equivariant classical features. Our hybrid architecture lifts each image into its eight-view D_4 orbit, processes all views with a shared morphology-aware MBConv encoder, and maps the resulting features to four eight-qubit quantum heads through learned angular encoding. Symmetry-tied single- and two-qubit operations, repeated data encoding, and orbit-averaged Pauli measurements produce an invariant representation. We compare this quantum branch with a parameter-matched classical orbit control while holding the encoder, data partitions, and model-selection protocol fixed. Across three simulated lensing regimes containing cold dark matter, axion-like dark matter, and no-substructure classes, the hybrid model achieves three-seed mean test accuracies of 0.9855, 0.9987, and 0.9998. These scores exceed the classical control by 2.42, 1.41, and 0.85 percentage points, respectively, while also yielding lower negative log-likelihood on every dataset. The findings indicate that symmetry-structured quantum state evolution can measurably refine a strong equivariant classical representation. They establish a controlled benchmark for hybrid quantum learning in scientific imaging without constituting a claim of quantum advantage.

Introduction

Dark matter is central to cosmology, yet its microscopic nature remains unknown. Cold and ultralight axion dark matter predict different small-scale structure (Bertone and Hooper 2018; Niemeyer 2020), which can perturb the arcs and Einstein rings produced by galaxy–galaxy strong lensing (Dalal and Kochanek 2002; Vegetti and Koopmans 2009). Because these weak signatures are entangled with source morphology, lens structure, resolution, and noise, simulation-based

machine learning has become an important analysis tool (Brehmer et al. 2019; Alexander et al. 2020; Varma, Fairbairn, and Figueroa 2020).

Lens classes are unchanged by the rotations and reflections forming D_4 on a square grid. Unlike standard convolutions, group-equivariant models enforce this symmetry through explicit group actions and shared parameters (Cohen and Welling 2016). We investigate whether a parameterized quantum circuit can further refine such a representation. Following learned classical-to-quantum interfaces (Wang, Theobalt, and Golyanik 2025), classical features determine bounded rotation angles, while repeated encoding expands the input-dependent function space without additional qubits (Schuld, Sweke, and Meyer 2021; Pérez-Salinas et al. 2020).

We process all eight D_4 -transformed views with fixed morphology channels and a shared MBConv encoder. Four eight-qubit heads use repeated encoding, shared rotations, and symmetry-tied interactions; orbit-reduced measurements yield invariant features (Meyer et al. 2023; Nguyen et al. 2024). Each qubit indexes a group element rather than an image pixel, so transforming an input permutes the circuit wires while the invariant readout removes orientation dependence. The symmetry is therefore built into the representation instead of being learned only through data augmentation. The central question is whether quantum state evolution adds useful capacity after the dominant classical feature extraction and symmetry constraints have already been controlled. A parameter-matched classical control therefore retains the same preprocessing, orbit construction, shared encoder, data partitions, and model-selection procedure, but replaces the quantum branch with orbit averaging. This design supports a direct module-level comparison while avoiding assumptions of quantum advantage.

Dataset

We use three simulated galaxy–galaxy strong-lensing datasets generated with `lenstronomy` (Birrer and Amara 2018). Each image contains an elliptical Sérsic source lensed by a $10^{12} M_\odot$ sheared singular isothermal elliptical halo. The three classes are cold dark matter with 10^6 – $10^{10} M_\odot$ point-mass subhalos, ultralight axion dark matter with vortex-like perturbations, and a smooth no-substructure reference. Axion masses are sampled between 10^{-24} and 10^{-22} eV (Toomey and collaborators 2026).

Model I contains 150×150 -pixel images with a 0.05 arc-sec pixel scale, Gaussian point-spread function, background and Poisson noise, and a reported signal-to-noise ratio of approximately 25. Models II and III contain 64×64 -pixel images and are described as Euclid-like and HST-like. These labels refer to the released configurations: the inspected generator does not save the computed Euclid noise and contains no separate HST instrument branch (Toomey and collaborators 2026).

The usable development sets contain 87,525, 89,104, and 89,396 images, with a fixed stratified 80:20 training-validation split. Each separate 15,000-image test set is reserved for final evaluation. We discard stored axion-mass metadata, replace non-finite pixels with zero, normalize images when needed, and resize them to 96×96 using anti-aliased bilinear interpolation.

Classical-to-Quantum Preprocessing

After removing non-finite values, normalizing the intensity, and resizing each image to 96×96 , we construct its complete D_4 orbit,

$$I_g = T_g I, \quad g = r^k s^f, \quad k \in \{0, 1, 2, 3\}, \quad f \in \{0, 1\}.$$

For every rotated or reflected view, a deterministic eight-channel morphology map $\mathcal{M}$ collects normalized and logarithmic intensity, Sobel-gradient magnitude, Laplacian response, 3×3 versus 9×9 multiscale contrast, radial and tangential derivatives, and the stabilized mixed derivative

$$\tanh \left| \partial_{xy}^S \left[\log \left(\frac{1 + 10^{-3}}{I_g + 10^{-3}} \right) \right] \right|^2.$$

The same MBConv encoder and angle projection are applied to all eight views:

$$\mathbf{z}_g = E_\phi(\mathcal{M}(I_g)) \in \mathbb{R}^{128},$$

$$a_{p,c,g} = \pi \tanh(\mathbf{w}_{p,c}^T \mathbf{z}_g + b_{p,c}).$$

Here $p \in \{1, \dots, 4\}$ indexes four quantum heads and $c \in \{Y, Z\}$. Each head uses eight qubits indexed by the elements of D_4 , rather than by image pixels. The angles control R_Y and R_Z gates in two data-reuploading layers, followed by shared rotations and symmetry-tied R_{ZZ} and R_{XX} interactions.

A transformed input only permutes the orbit representation,

$$\mathbf{z}_g(T_h I) = \mathbf{z}_{gh}(I),$$

while orbit-averaged X - and Z -basis measurements convert this equivariance into an invariant 48-dimensional vector, $\mathbf{q}(T_h I) = \mathbf{q}(I)$, for final classification.

Orbit Encoder and Quantum Circuit

D_4 -equivariant orbit encoder. The dark matter class of a lens is unchanged when its image is rotated by a multiple of 90° or reflected. A standard convolutional encoder provides translation equivariance, but it does not enforce a consistent

representation under these transformations. Data augmentation supplies transformed examples, but still allows the network to learn the same weak substructure feature separately at different orientations. We therefore construct the complete D_4 orbit

$$D_4 = \{e, r, r^2, r^3, s, rs, r^2s, r^3s\}$$

and process all eight views with the same morphology map and MBConv encoder. The encoder uses a 5×5 stem, eight MBConv blocks with depthwise convolutions and squeeze-excitation, and a final 1×1 convolution followed by global averaging. For each group element g , it produces

$$\mathbf{z}_g = E_\phi(\mathcal{M}(T_g x)) \in \mathbb{R}^{128},$$

$$\Theta_{a, :, g} = \pi \tanh([W_p \mathbf{z}_g + \mathbf{b}_p]_{a, :}).$$

The shared weights ensure that transforming the input only permutes the orbit features, $\mathbf{z}_g(T_h x) = \mathbf{z}_{gh}(x)$. The resulting angle tensor has shape $B \times 4 \times 2 \times 8$, containing two angles for every qubit in each of four quantum heads.

Quantum circuit. Each head is an independent eight-qubit circuit whose wires correspond to the eight elements of D_4 ; a qubit represents a transformed image view rather than a pixel. For head a , re-upload layer ℓ , and qubit g , the effective encoding angles are

$$\alpha_{a\ell g}^P = \lambda_{a\ell}^P \Theta_{a, g}^P + b_{a\ell}^P, \quad P \in \{Y, Z\}.$$

The circuit first applies $R_Y(\alpha_{a\ell g}^Y)$ and $R_Z(\alpha_{a\ell g}^Z)$, followed by shared trainable R_X , R_Y , and R_Z rotations. Information between related views is mixed using R_{ZZ} and R_{XX} gates tied over the rotation edges $E_r = \{\{g, rg\}\}$ and reflection edges $E_s = \{\{g, sg\}\}$ of the D_4 Cayley graph. This parameter sharing makes the circuit equivariant to the same permutation of qubits induced by the image transformation.

The complete block is applied twice using the same input features but independent trainable parameters. Each head and layer contains four encoding parameters, three shared single-qubit parameters, and four interaction parameters, giving

$$4 \text{ heads} \times 2 \text{ re-uploads} \times 11 = 88$$

trainable circuit parameters. The circuit is evaluated with differentiable TorchQuantum statevector simulation.

Invariant readout. From the final statevector, we compute local X and Z expectation values and XX and ZZ correlations over the rotation, half-turn, and reflection edge families E_r , E_{r^2} , and E_s . The half-turn edges are used only during readout. Orbit means, variances, edge-averaged correlations, and connected correlations give twelve invariant quantities per head and 48 features in total. These features are passed to a compact $48 \rightarrow 32 \rightarrow 3$ classifier.

Results and Ablation Analysis

We evaluate the hybrid architecture and its classical control on each separate official test set of 15,000 class-balanced images. Test images are excluded from training, validation, and checkpoint selection. We use three random seeds. For each seed, both networks are independently initialized and

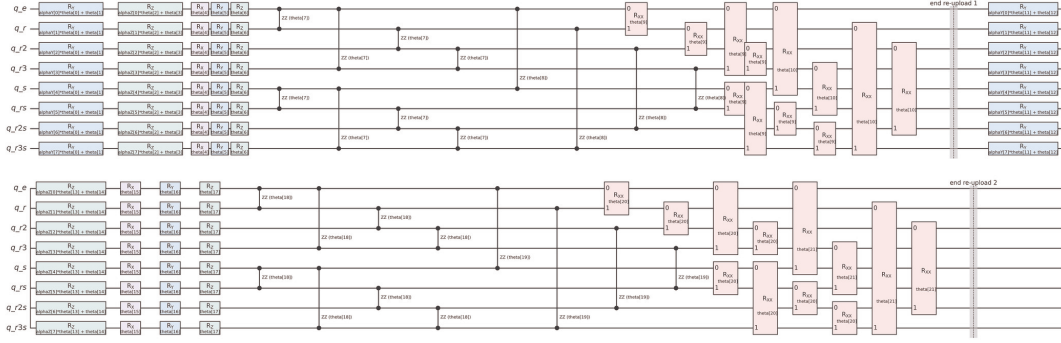


Figure 1: Decomposed circuit for one of the four quantum heads. The eight wires correspond to the elements of D_4 . Each re-upload contains angle encoding, shared single-qubit rotations, and parameter-tied R_{ZZ} and R_{XX} interactions. The two rows form one continuous circuit; measurements are omitted.

Data	Method	Mean acc. ± SD	Macro F_1 ± SD	NLL ± SD
Model I	Classical ctrl.	0.96127 ± 0.00182	0.96117 ± 0.00191	0.10833 ± 0.00487
	Hybrid	0.98547 ± 0.00112	0.98545 ± 0.00109	0.05317 ± 0.00268
Model II	Classical ctrl.	0.98460 ± 0.00094	0.98458 ± 0.00096	0.07022 ± 0.00315
	Hybrid	0.99873 ± 0.00022	0.99873 ± 0.00023	0.01747 ± 0.00111
Model III	Classical ctrl.	0.99133 ± 0.00071	0.99132 ± 0.00074	0.05437 ± 0.00277
	Hybrid	0.99980 ± 0.00008	0.99980 ± 0.00008	0.01321 ± 0.00069

Table 1: Official-test mean ± SD over three seeds; bold denotes the better method.

Data	Axion	CDM	No-sub.	Macro
Model I	0.9986	0.9973	0.9999	0.9986
Model II	0.9996	0.9998	1.0000	0.9998
Model III	1.0000	1.0000	1.0000	1.0000

Table 2: Hybrid one-versus-rest ROC–AUC on the official test sets.

trained end-to-end from scratch; reported scores are means over the three runs.

The hybrid architecture improves accuracy over the classical control by absolute margins of 0.02420, 0.01413, and 0.00847 on Models I–III, respectively. Macro F_1 follows the same trend, showing that the difference is not driven by a single class. The hybrid model also obtains a lower negative log-likelihood on all three datasets. Since every test set is class-balanced, accuracy and balanced accuracy coincide.

Figure 2 shows that the no-substructure class is separated almost perfectly across the three datasets. Model I remains the most challenging setting, where most errors occur between axion and cold dark matter images. This is also reflected by the lower CDM ROC–AUC of 0.9972926. The class boundaries become increasingly well separated for Models II and III, whose ROC curves approach the upper-left corner.

Classical-to-quantum bottleneck. Most spatial and morphological processing is performed by the classical front end. The deterministic morphology bank exposes intensity, edge, curvature, multiscale, radial, and tangential structures,

while the shared MBConv encoder maps every transformed image to a 128-dimensional representation. For each quantum head, this representation is compressed to two bounded angles controlling the R_Y and R_Z gates. The quantum circuit therefore receives neither the raw lens image nor the complete classical feature vector.

The classical control already achieves accuracies between 0.96127 and 0.99133, indicating that the classical encoder performs most of the discriminative feature extraction. The quantum circuit acts on top of this representation by mixing the compressed features of the eight D_4 -related views through repeated encoding and symmetry-tied R_{ZZ} and R_{XX} interactions. Its contribution is therefore interpreted as a refinement of a strong classical representation rather than independent learning from raw pixels.

Module-level ablation. The classical control retains the same morphology bank, eight-view D_4 lifting, and shared MBConv encoder. It replaces the quantum-processing branch with an average over the eight orbit representations followed by a linear classifier. The classical and hybrid configurations contain 242,725 and 245,221 trainable parameters, respectively, differing by approximately 1.0%.

Replacing the quantum-processing branch with classical orbit averaging reduces accuracy and macro F_1 across the repeated experiments. The largest difference occurs on Model I, while the margin becomes smaller as both configurations approach ceiling-level performance on Models II and III. This is a module-level ablation because the learned projection, state evolution, measurements, and final readout are

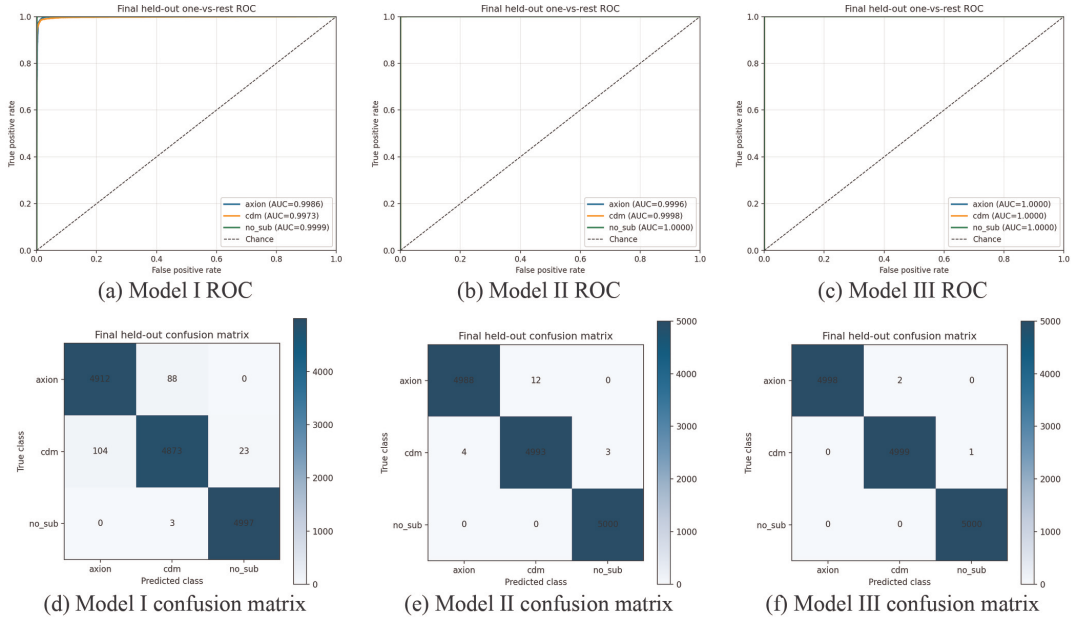


Figure 2: Official-test performance of the hybrid quantum-classical model. The upper row shows one-versus-rest ROC curves for the axion, cold dark matter, and no-substructure classes. The lower row shows the corresponding confusion matrices. All panels are generated from the validation-selected checkpoint evaluated on the separate official test set.

replaced together. The result characterizes the contribution of the complete symmetry-structured quantum branch and should not be interpreted as evidence of quantum advantage or as an improvement caused by a particular quantum gate.

Conclusion

We introduced a D_4 -equivariant hybrid quantum-classical architecture whose group-indexed qubits and tied circuit operations preserve the invariance of strong-lens classification. Across three datasets and random seeds, the hybrid model improves accuracy, macro F_1 , and negative log-likelihood over a closely sized classical control. The results support symmetry-aware quantum processing as a refinement of strong classical features, but neither isolate individual quantum components nor establish quantum advantage. Future work should test hardware noise, broader lens populations, and stronger classical orbit-mixing baselines.

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